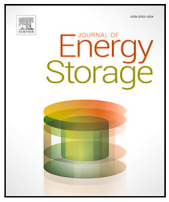
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Defining long duration energy storage

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ABSTRACT

Accelerating the deployment of variable renewable energy is changing the operational characteristics of the electric grid and creating an emerging need for storage technologies with extended energy-duration capabilities to maintain grid reliability. Extended or "long-duration" energy storage scales for supporting future grids are not well defined but can be anticipated to grow by estimating gaps in renewable performance of a decarbonized grid. Variable generation resources create a mismatch between electricity generation and use; as the amount of variable generation on the grid grows, so too do mismatches. This study reviews current uses of energy storage and how those uses are changing in response to emerging grid needs, then assesses how the power generation industry and academia are defining long-duration storage and organizing research efforts to develop commercial technologies. Using an illustrative example of a decarbonized grid, the study identifies the depth and breadth of future energy mismatches and concludes that two classes of long-duration energy storage will be needed in a decarbonized grid; one class lasting up to 20 h to manage daily cycles and one lasting for weeks or months to manage seasonal cycles. This study elucidates the necessity of long-duration energy storage in a decarbonized grid and may inform long-term planning processes. Grid planners can play an important role in the development of long-duration energy storage technologies through granular identification of storage needs that creates a market signal for investment in and development of the necessary technologies to provide a reliable and resilient grid for the future.

1. Introduction

In response to the growing impacts of climate change, government entities, utilities and private companies are adopting increasingly aggressive targets to deploy clean and renewable energy resources.

As the amount of variable, renewable generation on the grid increases, a grid operator's task of matching generation and demand in real time becomes more challenging. Changing customer demands, a changing climate, weather dependent energy, and increasingly severe weather events further exacerbate that challenge.

These emerging grid conditions are creating an imperative for long-duration energy storage (LDES) technologies to ensure supply availability, reconcile variable generation resources with uncertain customer demands, and strengthen the electric grid against weather events. There is a suite of technologies and treatments that can contribute to these objectives: energy efficiency, demand response and other forms of load flexibility, and the over-development of renewable energy resources to ensure minimum energy adequacy are just a few options. Emerging generation technologies such as geothermal, offshore wind, wave, and next-generation nuclear facilities may further contribute to decarbonization and grid stability.

This study will demonstrate that with currently available generation options for achieving grid decarbonization goals, the depth and breadth of mismatches between when electricity is generated and when it is consumed are simply too large to manage without the support of LDES technologies.

While LDES technologies are becoming a priority for grid operators and research organizations to fill these gaps, the definitions are general and typically express a minimum duration level. We offer a brief survey of existing definitions in use and then a proposal to target technology performance toward the specific characteristics of duration gaps.

The study concludes with an analysis to quantify the amplitude and frequency of energy generation surpluses and deficits in a decarbonized grid. This analysis demonstrates that the grid will need two types of LDES: one type that manages daily cycles and provides up to 20 h of storage, and another that manages seasonal cycles and provides storage measured in days or weeks.

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Table 1
Energy storage variables.

Symbol	Quantity	Definition
E	Energy	Rated discharge for time (MWh)
t	time	Duration of discharge (h)
P	Power	Rated output (MW)

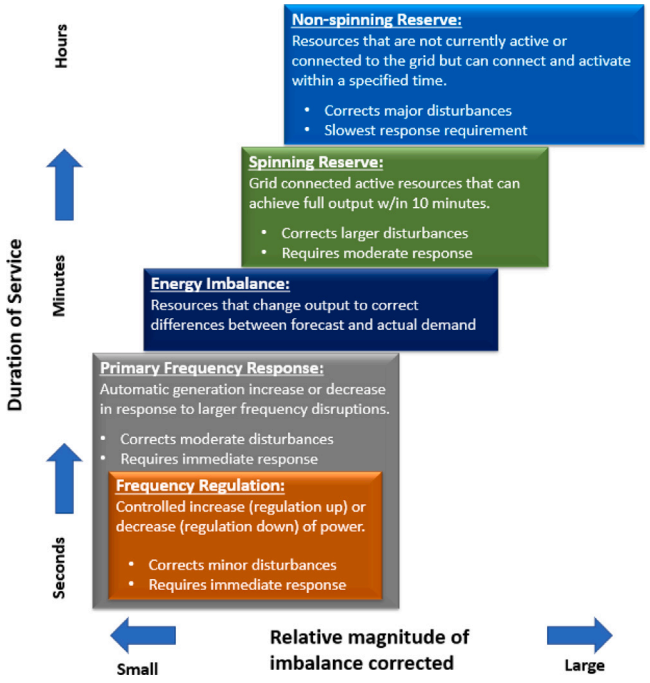


Fig. 1. Relative duration and magnitude of electrical ancillary services.

2. Measuring energy storage

Energy storage is a dispatchable source of electricity, which in broad terms this means it can be turned on and off as demand necessitates. But energy storage technologies are also energy limited, which means that unlike a generation resource that can continue producing as long as it is connected to its fuel source, a storage device can only operate on its stored energy or charge and once depleted, must then recharge before providing service again.

Because of this characteristic, energy storage technologies are measured in two dimensions. Their power rating represents the maximum amount of electricity they can discharge at a point in time and is generally measured in units of kilowatts (kW) and megawatts (MW). Their energy rating represents the total amount of energy that they can discharge before needing to recharge, which is generally measured in units of kilowatt-hours (kWh) and megawatt-hours (MWh).

While energy storage technologies are often defined in terms of duration (i.e., a four-hour battery), a system's duration varies at the rate at which it is discharged. A system rated at 1 MW/4 MWh, for example, may only last for four hours or fewer when discharged at its maximum power rating. But if it were able to be efficiently discharged at 0.5 MW, it would take about eight hours to fully discharge [1].

For the purposes of this study, duration will be defined as the length of time over which a storage technology can sustain its full rated power output, as expressed in Table 1.

$$Energy(E) = Power(P) * time(t) \quad (1)$$

3. Applications of energy storage

Energy storage systems provide a variety of services to ensure grid reliability. The duration of these services vary from milliseconds to potentially days or weeks. Ancillary services—those services necessary to correct short-term imbalances between generation and load—can be readily provided by energy storage technologies with a duration of two hours or less. Fig. 1 summarizes ancillary services in terms of the relative size of the imbalance they are designed to correct (x-axis) and the duration they require (y-axis).

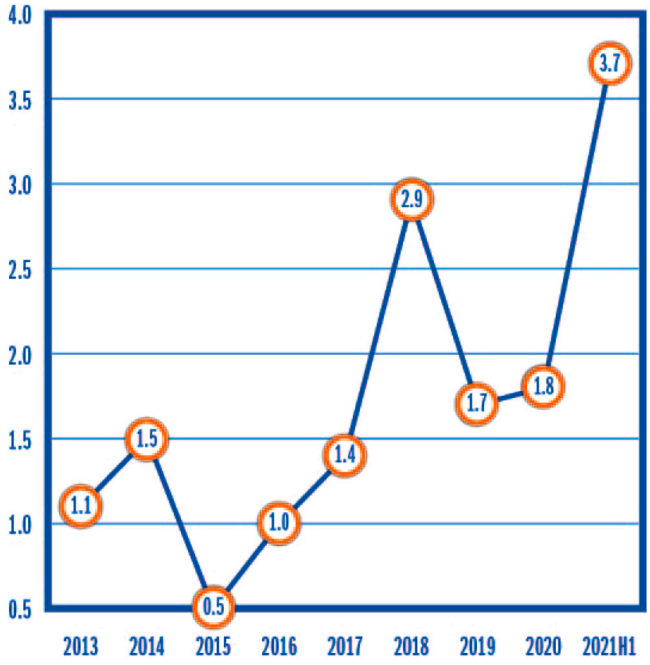


Fig. 2. Average duration of new utility-scale energy storage systems deployed in the U.S.,2013–2021 (hours). Chart data derived from the U.S. Energy Storage Monitor Q3 2021 report by Wood Mackenzie and the Energy Storage Association.

However, ancillary service markets are only a fraction of the size of energy and capacity markets, meaning that they offer comparatively less opportunity for resources to participate. According to data collected by the U.S. Energy Information Administration, 59 percent of energy storage assets deployed in the U.S. since 2016 are providing frequency regulation services. But 2020 also saw a dramatic increase in the share of storage assets also providing longer-duration services, such as spinning reserve, arbitrage, and load following, each of which grew by a factor of at least 3 from 2019 to 2020, to 39 percent, 37 percent, and 26 percent respectively. The share of systems providing peak shaving also slightly increased to 15 percent [2].

Corresponding with this shift in usage, the average duration of new energy storage systems has grown since 2013. While the average duration of new systems in 2013 was about one hour, the average duration of new systems deployed in the first half of 2021 was about 3.7 h, as shown in Fig. 2.

At longer durations, energy storage technologies can move beyond ancillary services and further contribute to grid reliability by providing sustained discharge during periods of high demand. Whether this service is compensated through energy market prices (arbitrage) or some capacity market mechanism, energy storage used in this manner helps maintain grid reliability by moving generation from low-demand periods to high-demand periods.

Fig. 3 uses observed load and generation data from the California Independent System Operator (CAISO) to illustrate the various grid services that energy storage can provide, particularly as the amount of variable generation on the grid increases. At the beginning of the day, there is a morning peak period from about 5 a.m. to 8 a.m. which drives up total demand (red line).

Customer demand is roughly predictable day to day. Grid operators prepare a day-ahead forecast to identify how much generation is

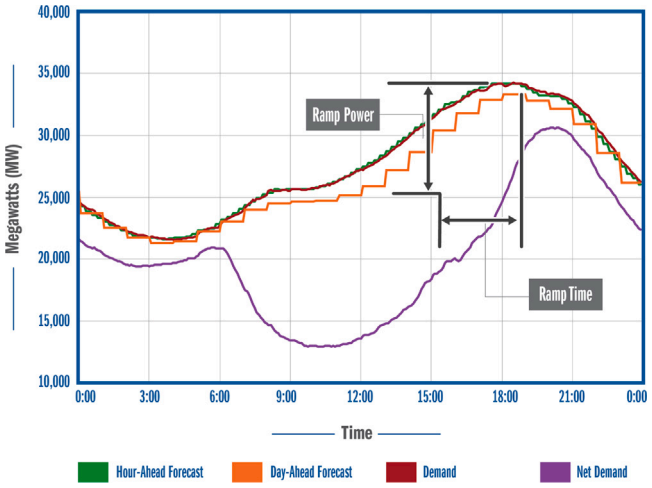


Fig. 3. Typical "Duck Curve" showing storage roles and daily operations. Note the two peaks, in the morning and evening. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

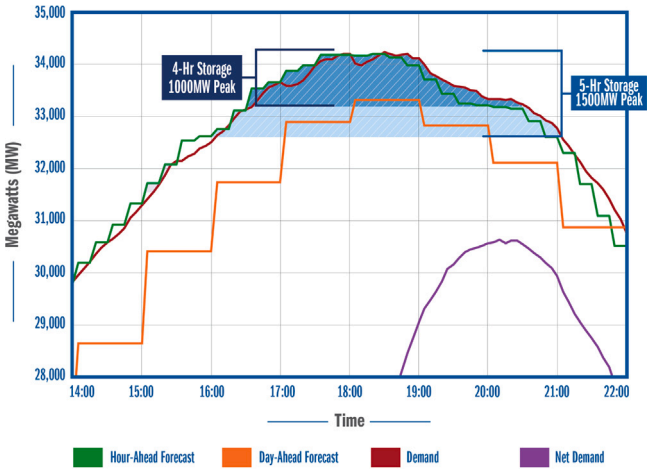


Fig. 4. Peak of typical "Duck Curve" showing how changes in storage duration result in larger capacity of storage. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

needed on an hourly basis (orange line) and procure that generation through the day-ahead market. CAISO also operates an hour-ahead market, which prepares more detailed forecasts of the following hour's energy demand in 15-min increments (green line) and procures additional generation as needed. As forecasts are never completely accurate, dispatchable energy sources are needed in real-time to correct the resulting imbalances between forecast and actual demand: CAISO primarily relies on the Western Energy Imbalance Market to provide this service. On a real-time basis, additional flexibility is procured in ancillary service markets, which select resources that can quickly respond to moment-to-moment changes in generation and demand to maintain them in balance and preserve system frequency.

As the day continues, approximately 12 GW of customer-sited solar resources begin producing energy, offsetting the amount of energy that customers need from the grid and resulting in the net demand curve (purple), which represents the actual demand that CAISO must meet.

Demand peaks at approximately 6 p.m., but from 5 p.m. to 8 p.m. the solar energy output decreases sharply to zero as the sun sets. With solar dropping and net demand increasing, ramping capacity—the ability to quickly bring resources online to both replace the diminishing solar generation and meet the rising demand—is essential. CAISO requires sufficient energy resources to ramp and replace the 12 GW of solar energy over a four-hour period as the sun sets. In addition to these resources, California pays energy providers for approximately 1.5 GW of a ramping product, to incentivize generator availability to accommodate unexpected fluctuations (decreases) in renewable output or sharp increases in demand.

The peak period for net demand lasts for about six hours, from 6 p.m. until midnight. Ensuring resource adequacy means that grid operators procure sufficient generation capable of sustained output through that peak period. At longer durations, energy storage resources can help provide resource adequacy. However, as more energy storage assets are dedicated to this service, the duration requirements increase as the remaining peak becomes flatter and longer, as illustrated in Fig. 4.

As peak periods get flatter and longer, incremental resource adequacy assets must meet longer duration requirements, or may have their capacity value pro-rated. For example, a four-hour battery deployed to provide resource adequacy for a six-hour peak period would only be credited and compensated for 2/3 of its rated capacity. This trend indicates increasing value for longer-duration storage technologies. As renewables continue to provide a greater percentage of the grid, longer-duration energy storage technologies will be necessary to meet future resource adequacy needs to accommodate the larger variation between generation (net demand) and load (demand) of any grid.

4. Existing long duration energy storage definitions

While the energy industry has yet to arrive at a standard definition, there is an emerging consensus that LDES means at least 10 h, which is summarized in Table 2.

(A) **DOE Storage Shot** The Department of Energy (DOE) Energy Earthshot initiative aims to accelerate development of grid-scale energy storage through reducing LDES costs. Specifically, the initiative seeks to deliver 10+ hours of storage within the next 10 years. Funding levels supporting the 10+ hours of storage are at $1.16 billion [3].

(B) **CPUC Integrated Resource Plan** The California Public Utilities Commission integrated resource plan rules require all regulated utilities to develop integrated resource plans to meet a reduced greenhouse gas emission target of the electric sector of 46 million metric tons in 2030. Within the categories, CPUC differentiates long and short duration storage. Long-duration storage is categorized as 8–12 h, where short-duration storage is categorized as 4 h or less.

(C) **California Energy Commission** The California Energy Commission in 2020 launched its own funding initiative for new LDES technologies, which it defined as 10+ hours [4]. Additionally, the CEC initiated a Grant Funding Opportunity (GFO) in 2020 to "…develop a clear understanding of the role that long duration energy storage (10 h or greater) can and should play in the future of California's grid and the optimal durations and location to support a variety of applications" [4].

(D) **ARPA-E DAYS Program** The Advanced Research Projects Agency-Energy (ARPA E) advances high-potential, high-impact energy technologies that are too early for private-sector investment. It funds research into long duration energy storage: the Duration Addition to electricitY Storage (DAYS) program is funding the development of 10 long duration energy storage technologies for 10–100 h with a goal of providing this storage at a cost of $.05 per kWh of output [5].

(E) **Long Duration Energy Storage Council** The Long Duration Energy Storage Council is a group of companies consisting of technology providers, energy providers, and end users whose focus is to replace fossil fuels with zero carbon energy storage to

Table 2
Long duration definition summary.

Organization	Duration
DOE storage shot	10+ h
CPUC IRP	8–12 h
CEC	10+ h
ARPA-E days program	10–100 h
LDES council	8–24 h
PJM	4,6,8,10 h

meet peak demand. In their report titled "Net-zero Power: Long Duration Energy Storage for a Renewable Grid", LDES is sorted into two archetypes based on duration the first is 8 to 24 h and the second is 24 h or more. The study examines the necessity for both durations and points out that these categories may serve roles for durations less than the limits as defined [6].

(F) **PJM Interconnection Capacity Requirements** PJM Interconnection, which is a transmission organization with electricity sales in multiple states was widely criticized for implementing a 10-hour duration requirement for valuing storage products participating in the capacity market. The method for calculating storage size required participants to claim capacity available to be dispatched over a full 10-h period, resulting in smaller ratings [7]. This requirement has since been revised into a series of classes which are 4, 6, 8, and 10 h.

5. Evolving long duration energy storage

Recent work in the research community indicates that where non-emitting sources of dispatchable generation, such as flexible nuclear and natural gas with carbon capture, are not available or not used, significant investment in storage technologies of 10 or more hours in duration will be necessary [8]. Shaner et al. [9] find that the necessity of LDES in a reliable power system specifically manifests in systems with greater than 80 percent variable renewable generation, as a means of integrating this variable generation across both daily and seasonal cycles. A fully decarbonized power systems with LDES of 100 h or more would likely cost less to operate than a system with shorter-duration storage technologies [10].

To develop a practical perspective on the value of and need for LDES in the U.S. electric grid, it is helpful to review the policies that are driving the grid's transition from dispatchable, greenhouse-gas emitting resources (such as coal and natural gas) towards non-dispatchable, non-emitting generation resources (such as wind and solar). Policies such as state-level renewable portfolio standards (RPS) coupled with falling technology costs drove the share of electricity generated by wind and solar from 0.2 percent in 2001 to 11.8 percent in 2020 [11].

Aggressive state decarbonization policies will accelerate this transition. To date, nine states, the District of Columbia, and Puerto Rico have adopted binding targets to achieve 100 percent clean energy generation. Two more states (Massachusetts and North Carolina) have binding standards to achieve net zero emissions in the electric sector. Another nine states have implemented nonbinding goals of 100 percent clean energy and 23 states (including some of those with voluntary 100 percent clean targets) have binding RPS targets of varying levels [12]. Additional demand for clean energy generation comes in the form of commitments by municipalities, utilities, and corporations. Finally, the federal government has set a nonbinding goal for all electric generation to be provided from clean resources by 2035.

Collectively, the binding clean energy targets set in 28 states and Washington D.C. will require about 34 percent of electricity consumed in the U.S. to be from non-emitting resources by 2050, based on those states' share of U.S. electricity consumption in 2020. In 2020, about 24 percent of the electricity generated in the U.S. was compliant with those standards, meaning that the amount of non-emitting generation on the grid will need to at increase by almost 50 percent by the year 2050.

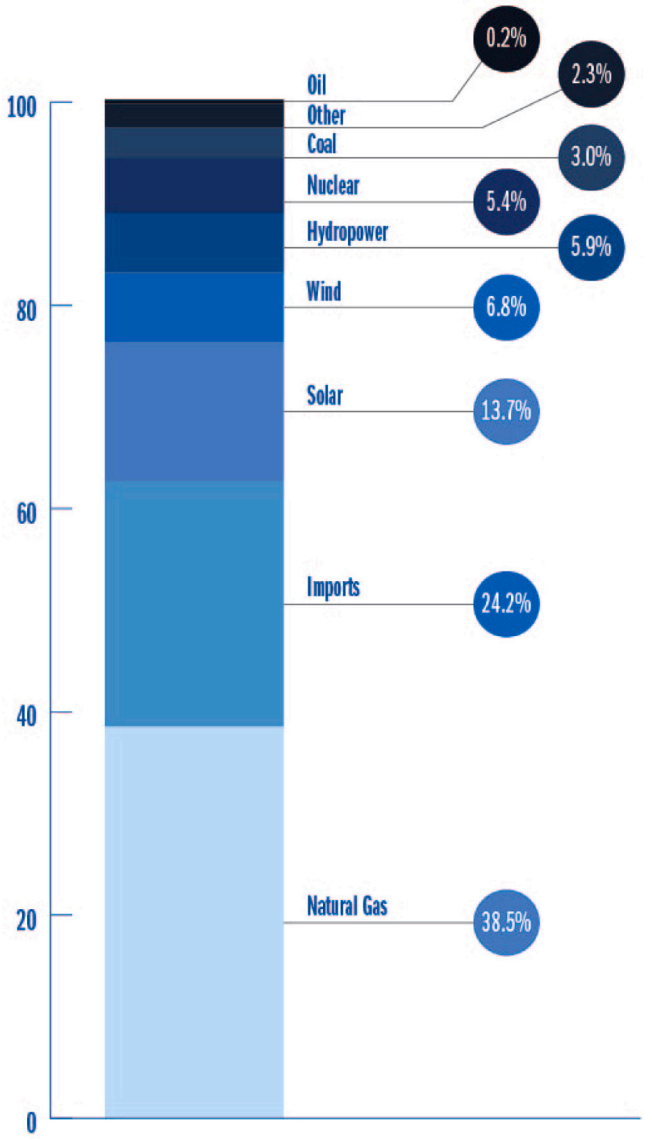


Fig. 5. California electric energy mix, August 2020 to July 2021.

Given the requirements of state policies and costs of available technology, it is likely that most of that new, eligible generation will come from variable generation sources such as wind and solar. To understand the impact of these technologies on grid operations, it is helpful to think of it as a matter of timing: variable generators create a mismatch between when electricity is generated and when it is used.

5.1. Illustrating the need for LDES in a decarbonized grid

To visualize how the mismatch between electric generation and consumption in a decarbonized grid will drive the need for LDES, this section presents a theoretical example based on a most recent year (August 2020–July 2021) of load and generation data for California, obtained from the Energy Information Administration's Hourly Electric Grid Monitor. This analysis is not meant to identify the most likely energy mix for the state in 2045, but rather to illustrate the transition to a carbon free power system shows no signs of being achievable without Long Duration Energy Storage due to potential deficits in generation.

Fig. 5 illustrates California's fuel mix from August 2020 through July 2021, which represents the starting point. California amended its RPS in 2018 to require 100 percent of the state's electricity to come from clean energy resources by 2045. As Fig. 5 shows, 41.7 percent of the electricity used in the state in the last year came from natural gas, coal and oil resources that will need to be replaced by 2045. Furthermore, the state's last nuclear plant is scheduled to retire in 2025, and no new nuclear facilities are planned. To achieve the state's mandate of 100 percent clean electricity by 2045, therefore, the state will have to replace 47.1 percent of its generation over the last year, with eligible generation resources.

The state's heavy reliance on imports is a more complicated matter. Over the course of the year, about 56 percent of imports came from the Pacific Northwest region and the other 44 percent came from the Southwest region. Both of those regions have multiple states (Washington and Oregon in the Pacific Northwest, Arizona and New Mexico in the Southwest) with 100 percent clean energy requirements themselves, and recent studies have cited changing load shapes and mounting capacity constraints in both regions [13,14]. Given those constraints, it is unlikely that California will be able to purchase firm capacity from those regions in the future in the quantities that it currently purchases them. This illustrative analysis therefore conservatively assumes that the state will also have to replace the 24.2 percent of the energy that it imported in 2020 with its own non-emitting generation resources, for a total of 71.3 percent of the energy that the state used in 2020 needing to be replaced by 2045.

This analysis removed the 71.3 percent of ineligible generation from the past year data and added sufficient wind and solar generation to replace the lost production on an annual basis. Two key assumptions were made in this process. First, given that California already has broad diversity in its wind and solar resources, it was assumed that the overall shape of the replacement wind and solar resources would follow the overall generation profile of the existing fleet. Second, wind and solar installations were increased by the same factor, to preserve the current ratio of generation (which is approximately 2:1 in favor of solar).

Increasing wind and solar generation each by a factor of 4.628 yields enough production to replace the 71.3 percent of generation that would be ineligible or unavailable in 2045.

This approach demonstrates that while enough energy would be generated throughout the course of the year to meet demand, there would be significant mismatches between when it is produced and when it is used. On an hourly basis, there would have been insufficient energy during 61 percent of hours in the year. The largest hourly deficit would have been 44.9 GW, while the largest hourly surplus would have been 60.9 GW. On a weekly basis, there would have been a net surplus in 23 weeks and a net deficit in 29 weeks.

Clearly, there are limitations to this simplified approach, which is meant to be an illustrative example of a fully decarbonized grid, rather than an authoritative prediction. There are multiple pathways to operating a decarbonized grid that would alter the imbalances identified here, such as overbuilding capacity or increasing regional coordination [15]. The deployment of new resource types like offshore wind could also affect the generation shape, while the growth of electric vehicles or increased electrification could affect the load shape. Additionally, while this analysis only added sufficient generation to equal demand, to the degree that storage is deployed to match supply and load, additional generation would be required to account for the round-trip losses of energy storage technologies. Recognizing those limits, the value of this study is in illustrating the depth and breadth of the generation deficits that would arise in a fully decarbonized system. While multiple strategies are available to incrementally adjust the size and timing of those deficits, they are simply too large to be overcome without LDES technologies in the portfolio.

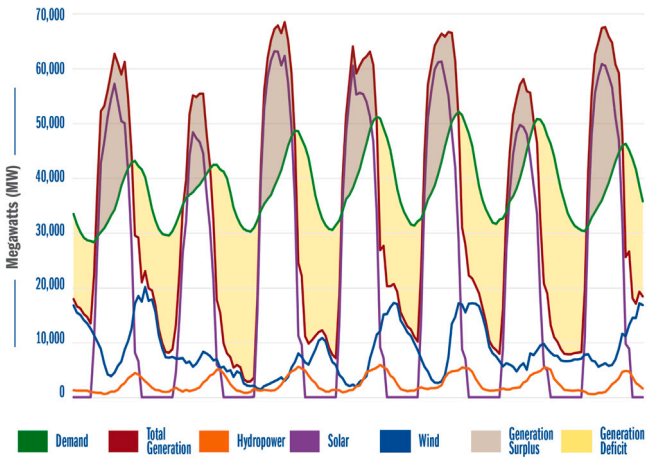


Fig. 6. Illustrative load and resource balance for California under full decarbonization, weekly resolution.

5.2. Future long-duration energy storage

Utilizing this illustrative example of a decarbonized grid, one can begin to organize the duration definition of LDES. The mismatches observed between generation and load in this example can be grouped into two primary classifications: diurnal, marked by daily cycles of excess and deficient generation, and seasonal, marked by sustained periods of net over- or under-generation on a weekly basis. These classifications point to two types of LDES: one for managing daily generation cycles, and one for managing seasonal cycles. This section will describe those needs in greater detail.

Fig. 6 illustrates the diurnal cycles of excess and insufficient generation over the course of the last week of July 2021 under this theoretical exercise.

As Fig. 6 demonstrates, California's solar-heavy grid generally results in excess generation during the day, while the smaller level of wind generation generally offsets some of the resulting evening and overnight deficits when the solar stops producing. Over the course of the year, overnight deficit periods last from 12 to 19 h, and average 14.6 h. This points to the first class of LDES: technologies with up to 20 h in duration that can manage those diurnal cycles.

Looking at the example from an annual perspective, weeks with a net surplus of generation run consecutively from mid-February until early August, though high loads coupled with below-average generation in the final two weeks of July 2021 would have resulted in net deficits. The net deficit weeks run consecutively from early August to mid-February. Fig. 7 summarizes the weekly net generation position over the course of the year.

Fig. 7 reveals the need for the second class of LDES: seasonal storage that can capture excess generation during seasons of surplus (spring to mid-summer in the case of California) and redistribute that energy over a period of several weeks to months during the seasons of deficit (mid-summer to winter in California).

Other studies have found that these cycles may be inverse in wind-heavy regions, which would likely be marked by sustained diurnal deficits during daylight hours and seasonal deficits during summer months [16]. Regardless of when the daily and seasonal mismatches between generation and load occur, the consistent finding across studies, which this study elucidates with an illustrative example, is that a deeply decarbonized system will require LDES to manage the daily and seasonal mismatches between energy generation and consumption and maintain system reliability.

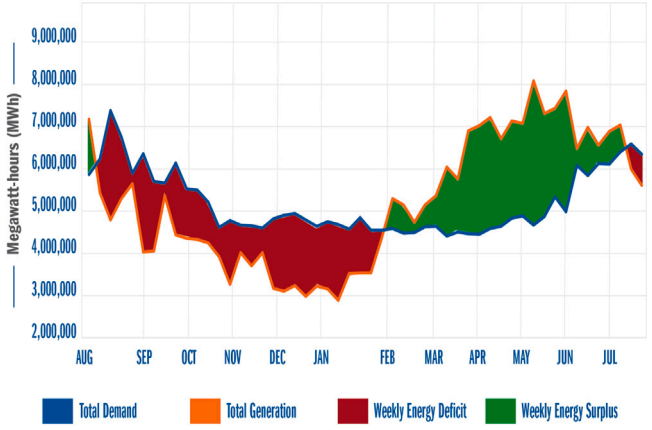


Fig. 7. Illustrative load and resource balance for California under full decarbonization, annual resolution.

6. Observed energy deficits

The section above considers an example of a future decarbonized grid to highlight the need for LDES resources. However, there are examples in existing power systems that also point towards this need, where energy shortfalls are being met from existing resources that may not be available in the future as the electric systems continue their path towards decarbonization.

In the United States, again taking the example of California, during August 14t and 15t, 2020, daytime solar output did not materialize relative to forecast, demand was high due to a heatwave, and imports were relatively unavailable because of high demand in the Northwest and wildfires limiting transmission availability. The system operator was left struggling to meet demand, which ultimately required power cuts. Even in a situation where forecasts were more accurate, additional resources would have been needed to meet solar output shortfalls.

Similarly, an analysis of wind output data across 2018 to 2020 for several U.S. organized markets and the Bonneville Power Authority balancing area shows that wind output varies significantly across days with some monthly variability evident as well. In one instance during July 2020, over a 72 h period, average hourly wind output in the PJM interconnection was less than 10% of the average hourly wind output for the month, representing a 3 GW shortfall. Although this may have been accurately forecast, large capacity resources would have been needed to address this shortfall if natural gas power plants were not available. This is despite this data being an aggregate of wind farms across the region.

Similar trends are observed internationally where renewable resources are at higher levels of deployment. Several European countries are seeing shortfalls in supply from variable generation output over daily and seasonal scales. In Central and South America, hydropower resources provide long duration water storage to manage resource variability. Costa Rica, for example, uses hydropower reservoir storage to manage seasonal dry periods of low water flow on its run-of-river hydroelectric system and short-term variability and longer-term seasonal droughts in its wind generation fleet.

7. Conclusion

This study has illustrated the necessity of LDES in achieving a reliable, decarbonized grid. While the magnitude and timing of LDES needs will depend on many factors, the nature of operating a grid that is predominantly powered by variable, renewable resources result in extensive mismatches between when energy is generated and when it is needed. Given the depth and breadth of the mismatches illustrated in this work, LDES technologies will be a necessary grid component to reconcile variable generation and variable demand.

In the energy and academic communities, there is an emerging consensus that "long" means at least 10 h of duration for LDES. This study has illustrated that two classes of LDES will be necessary in a decarbonized grid: one of up to 20 h of duration to manage daily cycles and another of seasonal duration to manage multi-month periods of generation deficits.

With an understanding of duration requirements, the technologies and economics of long duration energy storage need further consideration. Technologies to deliver LDES were not discussed here and aside from pumped hydroelectric and compressed air energy storage, other LDES technologies are not yet mature. The charging scheme of these LDES resources was not addressed in this study. Further, total generation will have to increase to supply adequate energy to fully charge the LDES. The economics of long duration versus multiple short-duration storage systems will also need further consideration.

In the meantime, it is important that grid planning processes begin to conduct more detailed modeling of decarbonized grid operations identifying the type, scale, and timing of LDES needs. Planning processes can play an important role in the development of those technologies by sending a market signal for research and development into LDES technologies to begin in earnest. To ensure that commercial technologies are available to meet those needs when they manifest, early investment signals are necessary. Commercially available resources, namely existing pumped storage and compressed air energy storage technologies, have geographic constraints that will limit their ability to meet the scale of LDES needs identified in this study.

CRediT authorship contribution statement

Jeremy Twitchell: Conceptualization, Draft preparation, Applying, Evolving, Potential decarbonized deficits. **Kyle DeSomber:** Submission coordinator, Draft preparation, Measuring, Applying and defining energy storage. **Dhruv Bhatnagar:** Draft preparation, Applying, Defining, Evolving energy storage.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

[1] P.J.M. Interconnection, Reliability Assurance Agreement Among Load Serving Entities in the PJM Region, Technical Report Rate Schedule FERC No. 44, 2011, URL: https://pjm.com/directory/merged-tariffs/raa.pdf.

[2] U.S. Energy Information Administration, Battery storage applications have shifted as more batteries are added to the U.S. grid, 2021, URL: https://www.eia.gov/todayinenergy/detail.php?id=50176.

[3] U.S. Department of Energy, Long duration storage shot, 2021, URL: https://www.energy.gov/eere/long-duration-storage-shot.

[4] California Energy Commission, GFO-19-308 - assessing long-duration energy storage deployment scenarios to meet California's energy goals, 2020, URL: https://www.energy.ca.gov/solicitations/2020-01/gfo-19-308-assessing-long-duration-energy-storage-deployment-scenarios-meet.

[5] ARPA-E, DAYS - arpa-e.energy.gov, 2018, URL: https://arpa-e.energy.gov/technologies/programs/days.

[6] Long Duration Energy Storage Council, McKinsey & Company, Net-Zero Power: Long Duration Energy Storage for a Renewable Grid, Technical Report, Long Duration Energy Storage Council, 2021, URL: https://www.ldescouncil.com/assets/LDES-2021-report-lowres.pdf.
[7] J. St. John, Taking aim at PJM's 10-hour duration capacity rule for energy storage, 2019, URL: https://www.greentechmedia.com/articles/read/taking-aim-at-pjms-10-hour-duration-capacity-rule-for-energy-storage.
[8] F.J.d. Sisternes, J.D. Jenkins, A. Botterud, The value of energy storage in decarbonizing the electricity sector, Appl. Energy 175 (2016) 368–379, http://dx.doi.org/10.1016/j.apenergy.2016.05.014, URL: https://www.sciencedirect.com/science/article/pii/S0306261916305967.
[9] M.R. Shaner, S.J. Davis, N.S. Lewis, K. Caldeira, Geophysical constraints on the reliability of solar and wind power in the United States, Energy Environ. Sci. 11 (4) (2018) 914–925, http://dx.doi.org/10.1039/C7EE03029K, Publisher: The Royal Society of Chemistry.
[10] N.A. Sepulveda, J.D. Jenkins, A. Edington, D.S. Mallapragada, R.K. Lester, The design space for long-duration energy storage in decarbonized power systems, Nat. Energy 6 (5) (2021) 506–516, http://dx.doi.org/10.1038/s41560-021-00796-8, URL: https://www.nature.com/articles/s41560-021-00796-8. Number: 5 Publisher: Nature Publishing Group.
[11] U.S. Energy Information Administration, Electricity data browser - net generation for all sectors, 2021, URL: https://www.eia.gov/electricity/data/browser/.
[12] National Conference of State Legislatures, State renewable portfolio standards and goals, 2021, URL: https://www.ncsl.org/research/energy/renewable-portfolio-standards.aspx.
[13] Z. Ming, A. Olson, G. De Moor, H. Jiang, N. Schlag, Long-Run Resource Adequacy under Deep Decarbonization Pathways for California, Technical Report, Energy and Environmental Economics, Inc., 2019, https://www.ethree.com/wp-content/uploads/2019/06/E3_Long_Run_Resource_Adequacy_CA_Deep-Decarbonization_Final.pdf.
[14] Public Service Company of New Mexico, 2020 Integrated resource plan, 2020, URL: https://www.pnmforwardtogether.com/irp.
[15] B.A. Frew, S. Becker, M.J. Dvorak, G.B. Andresen, M.Z. Jacobson, Flexibility mechanisms and pathways to a highly renewable US electricity future, Energy 101 (2016) 65–78, http://dx.doi.org/10.1016/j.energy.2016.01.079, URL: https://www.sciencedirect.com/science/article/pii/S0360544216300032.
[16] J.A. Dowling, K.Z. Rinaldi, T.H. Ruggles, S.J. Davis, M. Yuan, F. Tong, N.S. Lewis, K. Caldeira, Role of long-duration energy storage in variable renewable electricity systems, Joule 4 (2020) 1907–1928, http://dx.doi.org/10.1016/j.joule.2020.07.007, URL: http://www.sciencedirect.com/science/article/pii/S2542435120303251.

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